Tasfia Noor Chowdhury

Robot Learning from Demonstration · Biomedical AI & Wearable Sensing · Machine & Deep Learning
Dhaka, Bangladesh | tasfiaa.chowdhury@gmail.com | +880 1631 107299
GitHub | LinkedIn | ResearchGate | ORCID | Website

Education

Jul 2026 – Present (Expected Jun 2027)
M.Sc. in Robotics & Mechatronics Engineering
University of Dhaka (DU), Bangladesh
Thesis: *Knee Abnormality Detection from MRI Images* (RSNA dataset), MAIM Lab — Supervisor: Avishek Kumar Ghosh

Feb 2019 – Jun 2024
B.Sc. in Mechatronics & Industrial Engineering (4-year, with research thesis)
Chittagong University of Engineering and Technology (CUET), Bangladesh
CGPA: **3.63/4.00** | Senior-year GPA: **3.96/4.00** | Class rank: **8/30** | **Dean's Award ×5** (consecutive)

2016 – 2018
HSC GPA 5.00/5.00, Barisal Govt. Women's College | **SSC** GPA 5.00/5.00, Barisal Govt. Girls High School

Research Interests

- Robot Learning from Demonstration
- Imitation Learning & Human–Robot Interaction
- Machine Learning & Deep Learning
- Biomedical Engineering & Assistive Robotics
- Wearable Sensors & Biosignal Processing
- Medical Image Analysis & Computer Vision

Research Experience

Independent Researcher
Jul 2025 – Jun 2026
Remote

Robot Learning from Human Demonstration
Medical Robotics Lab, Friedrich-Alexander-Universität (FAU) Erlangen-Nürnberg, Germany — Mentor: M. A. Hafiz

- Webcam hand-gesture teleoperation of a dual Franka Panda in MuJoCo; expanded **one** demonstration into **50** physics-validated episodes (50/50 replay).
- Geometry-preserving augmentation lifted held-out behaviour-cloning success to **98% / 96%** vs. **48% / 26%** with naive noise (5 seeds, paired Wilcoxon p = 0.031).
- Learned warehouse navigation for TurtleBot3 (ROS 2 Jazzy, Webots) from teleoperation; a safety-aware rule layer cut close-obstacle exposure from **65% to 2.4%** of time, and the learned policy cut recoveries by **40.8%** (90 runs of 600 s).
- Outcome: two papers presented at IEEE RAAICON 2026.

M.Sc. Thesis
Jul 2026 – Present

Knee Abnormality Detection from MRI Images — MAIM Lab, University of Dhaka

- Developing deep-learning models to detect knee abnormalities from MRI scans (RSNA dataset).

B.Sc. Thesis
2023 – 2024

Hybrid Data-Driven Prediction of Inpatient Length of Stay — CUET, Supervisor: Kazi Naimur Rahman

- Big-data pipeline (Apache Spark on AWS) over **2.3 million** inpatient records; PCA, LightGBM, Random Forest, AdaBoost with cross-validation.
- Combined ML with process mining and simulation into a patient-flow framework. First-author preprint (arXiv 2025).

Research Project
2026

Generative Traffic Forecasting with Diffusion Models — ICML 2026 workshop poster (Forecast@ICML26)

- Reframed traffic forecasting as conditional denoising (DDPM, history-aware conditional UNet1D) on PeMSD7, with ensemble-medoid inference (N = 10); preserved sharp shockwaves and backward-moving phantom jams that MSE-trained baselines smooth out (medoid MAE **5.32 mph**).

Publications

Conference Papers

[C1] F. Shahriar, M. A. Hafiz, **T. N. Chowdhury**, W. N. Iqra, "**Hand-Gesture-Guided One-Demo-to-Many Robot Skill Generation with Teacher-Weighted Residual Correction in MuJoCo**," *2026 IEEE 5th Int. Conf. on Robotics, Automation, Artificial Intelligence and Internet of Things (RAAICON)*, 2026.

[C2] W. N. Iqra, M. A. Hafiz, F. Shahriar, **T. N. Chowdhury**, "**Human-Demonstration-Guided Warehouse Navigation with Safety-Aware Gap Reasoning for Mobile Robots**," *2026 IEEE 5th Int. Conf. on Robotics, Automation, Artificial Intelligence and Internet of Things (RAAICON)*, 2026.

Workshop Paper

[W1] M. I. Hoque, M. I. Khalil, **T. N. Chowdhury**, T. B. H. Jerin, "**Generative Traffic Forecasting: Preserving Shockwave Topology with Diffusion Models**," *ICML 2026 Workshop on Forecasting (Forecast@ICML26)*, **poster**, 2026. openreview.net/forum?id=KA5RdNascV

Journal Article

[J1] N. Sultana, A. M. Moosa, S. M. F. Faisal, **T. N. Chowdhury**, "**Machine Learning Models for Optimizing Online Order Fulfillment: Forecasting Lead Time and Late Delivery Risk**," *Journal of Optimization and Supply Chain Management*, 2(2), 70–81, 2025. doi:10.22034/ISS.2025.9373.1049

Preprints

[P1] **T. N. Chowdhury**, S. A. Mou, K. N. Rahman, "**A Hybrid Data-Driven Approach for Analyzing and Predicting Inpatient Length of Stay in Health Care Centers**," arXiv, 2025. arXiv:2501.18535

[P2] S. A. Mou, **T. N. Chowdhury**, A. I. Mannan, S. N. Mim, L. Tarannum, T. Noman, J. U. Ahamed, "**AI-Driven Water Segmentation with Deep Learning Models for Enhanced Flood Monitoring**," arXiv, 2025. doi:10.48550/arXiv.2501.08266

Teaching Experience

STEM Instructor
Jan 2021 – Dec 2025

Freelance Instructor & Academic Mentor (remote and on-site)

- Taught Physics, Mathematics and IT with customised curricula, adapting methods to each student's learning pace.

Projects

Award-winning 2025 — **EV Battery-Charging Stations for Electric Three-Wheelers** — Winner, Solve-a-thon 2025 (EU, World Bank, WePower)

Assistive Robotics 2022 — **Eye-Gaze Controlled Wheelchair** — 2nd Runner-Up, Project Idea Competition

Automation Course project — **Vision-Based Automated Box-Sorting Station** — PLC (Ladder Diagram) in Factory I/O with sensor–actuator synchronisation

Technical Skills

Programming	Python (NumPy, Pandas), C, MATLAB/Simulink
Machine Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM; U-Net, ResNet, DeepLab v3; behaviour cloning, MLP policies
Robotics & Simulation	ROS 2 (Jazzy, Nav2), MuJoCo, Webots, Gazebo; Franka Panda (sim), TurtleBot3, UR3 cobot; LiDAR, teleoperation
Data & Cloud	Apache Spark, AWS, Matplotlib, Seaborn, Plotly, Power BI
Automation & Hardware	Siemens TIA Portal, Factory I/O, Arduino, Proteus, SolidWorks, AutoCAD, 3D printing

Achievements & Awards

Winner, Solve-a-thon 2025 (EU, World Bank, WePower, Devtale Partners)	2025
Fellow, Aspire Leaders Program (Harvard Business School initiative)	2025
Graduate, Shoktikonna Leadership Cohort (EU, GIZ, World Bank)	2025
Dean's Award, five consecutive terms (GPA ≥ 3.75), CUET	2021–2024
Dutch-Bangla Bank Foundation Scholarship, full undergraduate stipend	2019–2024
2nd Runner-Up, Project Idea Competition (Eye-Gaze Controlled Wheelchair)	2022
Regional Winner, Physics Olympiad and English Olympiad	2017
Government Scholarships for results in JSC, SSC and HSC	2013–2018

Leadership & Service

Technical Lead
Jul 2023 – Present

Communication & Technical Lead, ILM International School (remote) — led communication and design workflows; enrolment up 25%.

Founder
2021 – Present

Noor Group (e-commerce) — Community Startup Award for exceptional growth, 2021.

References

Kazi Naimur Rahman
Assistant Professor, Dept. of MIE, CUET
B.Sc. thesis supervisor
naimur@cuet.ac.bd

Sanjeeb Roy
Assistant Professor, Dept. of MIE, CUET
(Doctoral student, Univ. of Arizona)
sanjeeb@cuet.ac.bd

Monowar Wadud Hridoy
Assistant Professor, Dept. of MIE, CUET
hridoy@cuet.ac.bd